

Group Coursework Submission Form (PA)

Specialist Masters Programme

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1 Introduction

We price and analyse a UniCredit variable-rate bond (ISIN IT0005599110) whose capped-and-floored coupon embeds interest-rate optionality. Using QuantLib, we decompose the bond into a leveraged FRN leg, a net option component, and a CVA adjustment, obtaining a model clean price of €1,055.11. We build a PCA-based factor model, compute VaR/ES analytically and via Monte Carlo, and design an IRS + CDS hedge that cuts VaR by 84.8%. We then compare our price with the EuroTLX quote of €1,017.00 and discuss the €38 gap.

2 Product Description and Market Data

2.1 Financial Product

The financial product under analysis is the UniCredit Variable Rate Bond 2034, a structured floating-rate note whose leveraged, collared coupon creates non-linear interest-rate optionality [4, Pt. I]. Bond characteristics are presented in Table 1.

Field	Value	Field	Value
ISIN	IT0005599110	Issue Date	12 June 2024
Issuer	UniCredit S.p.A.	Maturity	12 June 2034
Programme	€60bn EMTN	Tenor	10 years
Currency	EUR	Status	Senior (MREL-eligible)
Nominal	€20,000,000	Credit Rating	A2 (Moody's equiv.)
Denomination	€1,000 / note	Governing Law	Italian law
Issue Price	100% of par	Listing	EuroTLX; MOT
Series/Tranche	742 / 1	Clearing	Monte Titoli; Euroclear; Clearstream

Table 1: Bond characteristics. Source: UniCredit prospectus.

The bond pays back 100% of face value on 12 June 2034. The quarterly in-arrear coupons are given by

$$c(r) = \max(0\%, \min(1.60 \times r, 5.45\%)) \quad (1)$$

where r is the 3M EURIBOR fixing on the second TARGET settlement day before each period, day count 30/360, Modified Following Business Day (MFBD).

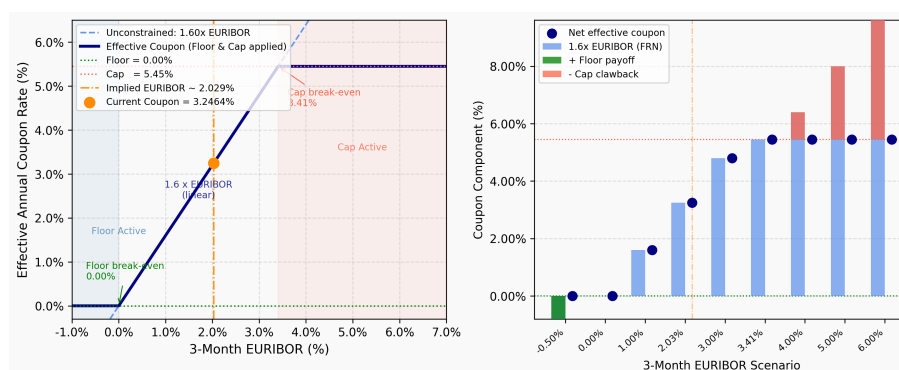


Figure 1: Coupon payoff profile and scenario-based component decomposition.

Three regimes can be defined: floored (0% , $r \leq 0\%$), linear ($1.60r$, $0\% < r < 3.406\%$), and capped (5.45% , $r \geq 3.406\%$) [7, Ch. 29]. The payoff is shown in Figure 1. The coupon formula in equation (1) can be written as:

$$c(r) = \underbrace{1.60 \times r}_{\text{Leveraged FRN leg}} + \underbrace{\max(0\% - 1.60r, 0)}_{\text{Long Floor @ 0\%}} - \underbrace{\max(1.60r - 5.45\%, 0)}_{\text{Short Cap @ 5.45\%}} \quad (2)$$

As shown in Table 2, the coupon is broken down into four simpler building blocks.

Additionally, **Replication B** (fixed bond + receiver swap + collar) [2, 7] overlay a receiver swap (receive $c_{\text{mid}} = 2.73\%$, pay $1.60r$) on a fixed-rate bond against a risk-free benchmark [2, Ch. 6]. **Replication C** (caplet bull spread) each coupon equals a long caplet at $K_1 = 0\%$ minus a short caplet at $K_2 = 3.406\%$, scaled by 1.60 [7, Ch. 20]. **Replication D** (digital approximation) "all-or-nothing" bet i.e. if r is 3.406% or above, the cap pays out a flat 5.45% otherwise it pays nothing estimated using Bachelier [7, Ch. 26].

#	Component	Instrument	Position / Strike
1	FRN leg	FRN (1.6 $\times$)	Long; 1.60 r per period
2	IR Floor	Floorlet strip	Long; strike 0%
3	IR Cap	Caplet strip	Short; strike 5.45%
4	Redemption	Zero-coupon bond	Long; 100% at maturity

Table 2: Replicating portfolio decomposition.

2.1.1 Historical Coupon Analysis

Figure 2 shows all coupons paid to-date using Bank of Finland (Suomen Pankki) 3M EURIBOR fixings.¹ We use $\alpha = 0.25$ (30/360) [2, App. B, 4, Ch. 2]. Coupons 1,2 were cap-binding ($r = 3.743\%$ and 3.460% , both above 3.406%) costing investors approximately €1.56/€1,000 in upside (€1.35 + €0.22). ECB rate cuts over 2024–2025 resulted in the reference rate falling ≈ 179 bps, to a trough of 1.954% (June 2025). By September–December 2025 the rate recovered to 2.029–2.082%, with coupons of 3.246–3.331% still ≈ 135 bps below the 3.406% cap break-even.

#	Reset Date	Period Start	Period End	Adj. End	EURIBOR (%)	Coupon (%)	Amount (€)	Status
1	10 Jun 2024	12 Jun 2024	12 Sep 2024	12 Sep 2024	3.743	5.4500†	13.625	Paid
2	10 Sep 2024	12 Sep 2024	12 Dec 2024	12 Dec 2024	3.46	5.4500†	13.625	Paid
3	10 Dec 2024	12 Dec 2024	12 Mar 2025	12 Mar 2025	2.872	4.5952	11.488	Paid
4	10 Mar 2025	12 Mar 2025	12 Jun 2025	12 Jun 2025	2.547	4.0752	10.188	Paid
5	10 Jun 2025	12 Jun 2025	12 Sep 2025	12 Sep 2025	1.954	3.1264	7.816	Paid
6	10 Sep 2025	12 Sep 2025	12 Dec 2025	12 Dec 2025	2.029	3.2464	8.116	Paid
7	10 Dec 2025	12 Dec 2025	12 Mar 2026	12 Mar 2026	2.082	3.3312	8.328	Paid
						Total:	73.186	

Figure 2: Coupon schedule per €1,000 face value; all dates and amounts generated via QuantLib.

2.1.2 Best- and Worst-Case Scenario Analysis

We evaluated scenarios by modifying the 3M reference rate but keeping the discount curve fixed:

- **Best case**— r stays above the cap break-even (3.406%) at every future reset, so every coupon pays the maximum 5.45% p.a.
- **Base case**— future reference rates are taken from the forward curve built in our bootstrapping step. We then calculate the coupon using the collar formula (floor at 0%, cap at 5.45%, multiplied by 1.60).
- **Worst case**— r equals 0% or below at every future reset and every coupon yields 0%. The bondholder will receive only the par amount at maturity.

Table 3 shows fair values spanning €819–1,222 ($\approx 40\%$ of par). The base case yields €1,119 (112% of par). Figure 3 illustrates the cash-flow profiles.

Scenario	Cpn PV	Rdm PV	Total PV	% Par
Best (cap)	420.11	802.26	1 222.37	122.2%
Base (fwd)	316.84	802.26	1 119.10	111.9%
Worst (floor)	16.37	802.26	818.63	81.9%

Table 3: PV decomposition (per €1,000 face).

2.2 Market Data and Decision-Making

2.2.1 Choice of Reference Rate and Data Sources

The term structure is based on EUR instruments observed on 5 November 2025 (source: LSEG Refinitiv, dataset MarketData202526.xlsx):

- **EURIBOR cash deposits** (overnight (O/N)–6M): unsecured interbank rates that yield model-free discount factors at the short end.
- **EUR IRS (AB6E, 1Y–10Y)**: mid par fixed rates vs. 6M EURIBOR (annual fixed 30/360, semi-annual float ACT/360). Bid-offer spreads of 0.1–0.3 bps make these the most liquid long-end EUR rate instruments [8, Ch. 2].

¹https://www.suomenpankki.fi/en/Statistics/interest-rates/charts/korot_kuviot/euriborkorot_pv_chrt_en/

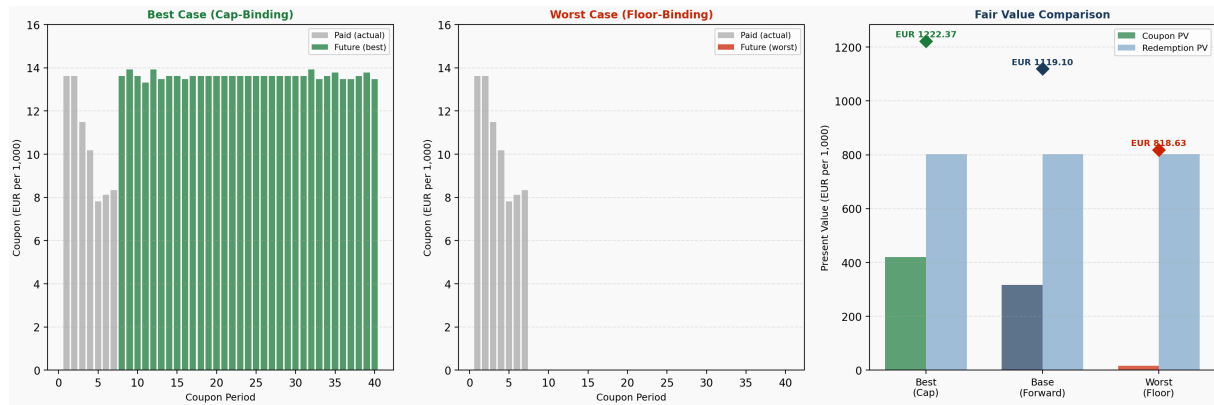


Figure 3: Scenario analysis: best/worst coupon cash flows and fair-value comparison.

2.2.2 Justification of Market Data Selection

We use short-term deposits (O/N–6M, settled T+2, ACT/360) because we can calculate a discount factor directly [3, p. 92]. We chose to drop the 12M deposit and instead use 1Y IRS because the swap has tighter bid–offer spreads and we can compute discount factor and forward rate in one go.

2.3 Methodologies

2.3.1 Bootstrapping Procedure

We use bootstrapping to build zero-coupon discount-factor curve [3, Ch. 3].

Short end: deposits. For a deposit with quoted ACT/360 rate r_d and maturity T :

$$DF(T) = \frac{1}{1 + r_d \cdot \alpha}, \quad \alpha = \frac{\text{actual days}}{360} \quad (3)$$

We use four deposits (1W, 1M, 3M, 6M) for maturities of 0.02–0.50 years.

Long end: IRS. At inception, fixed leg equals floating leg because a par swap has zero NPV:

$$c \sum_{i=1}^n \alpha_i^{\text{fix}} DF(t_i) = DF(t_0) - DF(T_n) \quad (4)$$

where t_0 is the settlement date and $\alpha_i^{\text{fix}} = 1$ (annual 30/360 for full coupon years). Rearranging for the unknown terminal discount factor:

$$DF(T_n) = \frac{DF(t_0) - c \sum_{i=1}^{n-1} \alpha_i^{\text{fix}} DF(t_i)}{1 + c \alpha_n^{\text{fix}}} \quad (5)$$

All prior $DF(t_i)$ are known and each new maturity adds exactly one equation and one unknown, so the system is solved sequentially without iteration [7, p. 437]. We implement in QuantLib via `ql.DepositRateHelper` and `ql.SwapRateHelper` combined in a `ql.PiecewiseLogLinearDiscount` term structure.

2.3.2 Interpolation Methods

For a date t between pillars T_1 and T_2 :

$$\ln DF(t) = \ln DF(T_1) + \frac{t - T_1}{T_2 - T_1} [\ln DF(T_2) - \ln DF(T_1)] \quad (6)$$

This is algebraically equivalent to assuming a *piecewise-constant instantaneous forward rate* $f_{[T_1, T_2]}$ between each pair of adjacent pillars:

$$f_{[T_1, T_2]} = \frac{\ln DF(T_1) - \ln DF(T_2)}{T_2 - T_1} \quad (7)$$

We chose Log-linear interpolation for three reasons [6]:

1. since DF is monotonically decreasing, $f_{[T_1, T_2]} > 0$ is guaranteed at all inter-pillar dates.
2. a change in one pillar quote affects only the two adjacent intervals, not the entire curve.
3. no root-finding between pillars; a single log-linear formula suffices.

The step-like pattern in the 3M forward curve (Figure 4, panel 2) is a result of the piecewise-constant forward assumption. A Monte Carlo comparison of term-structure estimation techniques by Biuno, Gregory-Allen, and Yaari [1] found that exponential-spline and piecewise-linear methods deliver similar accuracy at typical market data densities, justifying our selection to use simpler log-linear approach.

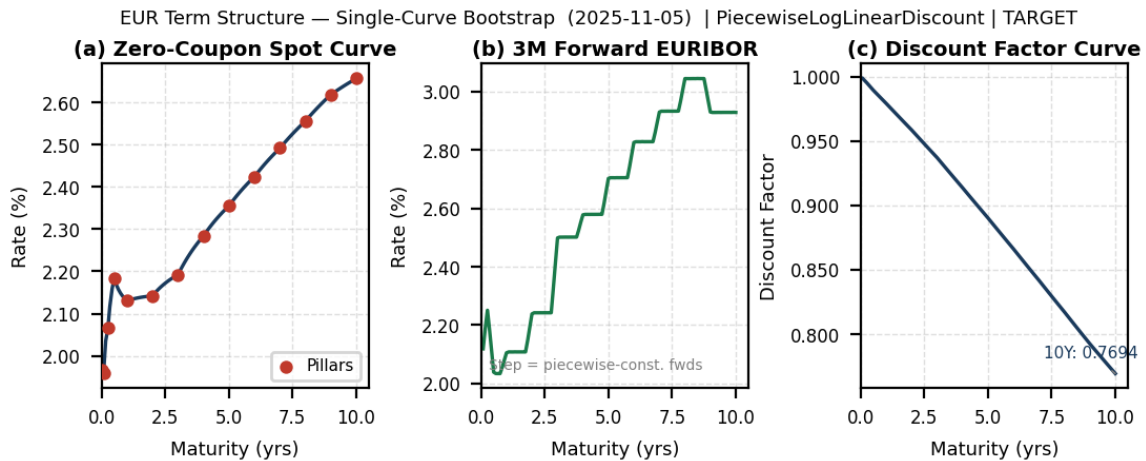


Figure 4: EUR term structure bootstrap (5 Nov 2025): (a) zero-coupon spot curve, (b) 3M forward EURIBOR, (c) discount factors.

2.3.3 Curve Construction and Validation

We observe the bootstrapped curve has 15 nodes and spans 0.01-10.25 years. Selected zero rates and discount factors are summarised in Table 4 and in Figure 4. We have validated that every input instrument reprices to within 0.0000 bps (Figure 5), the theoretical minimum for an exact single-curve bootstrap. The curve is upward-sloping with a mild positive convexity between 1Y and 10Y while longer maturities reflected a gradual return to neutral [8, Ch. 5].

3 Pricing Procedures and Model Justification

3.1 Pricing Procedures

3.1.1 Decomposition into Replicating Instruments

Using Equation (2), the option-like part of the coupon is valued as a strip of quarterly floorlets and caplets on 3M EURIBOR with effective notional $N^* = 1.60 \times N$:

$$c(r) = 1.60r + \max(0 - 1.60r, 0) - \max(1.60r - 5.45\%, 0) \quad (8)$$

Hence, for valuation at t_0 ,

$$PV_{\text{opt}}(t_0) = \sum_i PV(\text{floorlet}_i) - \sum_i PV(\text{caplet}_i), \quad (9)$$

with floor strike $K_F = 0\%$ and cap strike on EURIBOR $K_C = 5.45\%/1.60 = 3.40625\%$.

Maturity	DF	Zero (ann.%)	Fwd 3M (%)
1Y	0.9793	2.131	2.105
2Y	0.9586	2.142	2.238
3Y	0.9370	2.192	2.498
5Y	0.8902	2.355	2.702
7Y	0.8418	2.492	2.929
10Y	0.7694	2.657	2.928

Table 4: Bootstrapped zero rates and discount factors at key maturities (5 Nov 2025).

Instrument	Tenor	Input (%)	Implied (%)	Error (bps)
Deposit	1W	1.9210	1.9210	+0.0000
Deposit	1M	1.9140	1.9140	+0.0000
Deposit	3M	2.0230	2.0230	-0.0000
Deposit	6M	2.1420	2.1420	-0.0000
IRS (AB6E)	1Y	2.1300	2.1300	+0.0000
IRS (AB6E)	2Y	2.1420	2.1420	-0.0000
IRS (AB6E)	3Y	2.1930	2.1930	+0.0000
IRS (AB6E)	4Y	2.2815	2.2815	+0.0000
IRS (AB6E)	5Y	2.3498	2.3498	+0.0000
IRS (AB6E)	6Y	2.4155	2.4155	+0.0000
IRS (AB6E)	7Y	2.4803	2.4803	+0.0000
IRS (AB6E)	8Y	2.5398	2.5398	+0.0000
IRS (AB6E)	9Y	2.5975	2.5975	+0.0000
IRS (AB6E)	10Y	2.6330	2.6330	-0.0000

Figure 5: Bootstrap validation: input vs. curve-implied rates. Maximum absolute error: 0.0000 bps across all 14 instruments.

3.1.2 Discounting and Forward Rate Estimation

Discount factors and forwards are taken from the Q3 term structure (Section 2.3.1), built from EUR deposits and AB6E swaps on 5 Nov 2025. For each coupon period $[T_{i-1}, T_i]$ we use simple forward F_i from the curve and use $\alpha_i^{30/360}$.

3.1.3 Cap and Floor Pricing via Black-76 / Bachelier Model

We implement Bachelier (normal) model consistent with the market sheet `Bachelier` in `MarketData202526.xlsx`. For each optionlet with expiry τ_i , forward F_i , strike K , and normal volatility σ_i^N , the unit payoff price is:

$$V_i^{\text{Bach}} = \text{BachCall/Put}(F_i, K, \sigma_i^N \sqrt{\tau_i}), \quad (10)$$

and the cashflow PV is:

$$PV_i = N^* \alpha_i^{30/360} DF(t_0, T_i) V_i^{\text{Bach}}. \quad (11)$$

we then calibrate in two steps:

1. Read ATM 6M EUR cap quotes (maturities 1Y to 10Y) from the market workbook.
2. We extract the implied normal volatility for each individual caplet, one maturity at a time, by matching the cap's weighted variance

$$\sigma_{\text{cap}}(T_n)^2 A_n = \sum_{j=1}^n w_j \sigma_j^2, \quad w_j = \alpha_j^{\text{ACT}/360} DF(t_0, T_j), \quad (12)$$

3.2 Model Justification

3.2.1 Choice of Interest Rate Model

Since the coupon (Equation (1)) depends only on 3M EURIBOR caplet fixings, we price each caplet directly under the Bachelier (normal)(Section 3.1.3). We used discount factors from Section 2.3.1. We preferred this cap-floor approach over leveraged-collar structure because it is arbitrage-free and consistent with quoted market prices [7, Ch. 29].

Component	PV (€/1 000)
Long floorlet strip PV	3.0586
Short caplet strip PV	31.5886
Net option PV (floor – cap)	–28.5300

Table 5: Option component PV (5 Nov 2025).

4 Implementation of Pricing Procedure

4.1 Implementation Overview

4.1.1 Yield Curve Construction in QuantLib

We implement the bootstrapping system as in equations (3)–(5) using QuantLib via `PiecewiseLogLinearDiscount`. The curve construction is implemented in `Q3/q3_main.py`; Table 4 and Figure 4 present the resulting zero rates, 3M forward rates, and discount factors.

4.1.2 Coupon Schedule and Cash Flow Generation

Coupon dates are generated with QuantLib under TARGET calendar, Modified-Following convention, quarterly frequency, and 30/360 BondBasis day count. Reset dates are set to two TARGET business days before each period start. Only periods with reset date strictly after valuation date contribute option value in Q5.

4.1.3 Valuation of Embedded Options

Embedded options are valued in `Q5/q5_main.py`. The implementation:

1. loads ATM cap quotes from the Bachelier sheet,
2. strips caplet normal vols by sequential variance matching,
3. prices each caplet/floorlet with QuantLib's Bachelier formula,

4.1.4 CVA Computation

We compute CVA in `Q6/q6_main.py`. The implementation:

1. builds the 40-period coupon schedule and projects base-case cash flows using historical fixings (periods 1–7) and Q3 forward rates (periods 8–40), applying the leveraged-collar formula,
2. adds the €1,000 par redemption at maturity,
3. constructs a flat-hazard survival curve from the 5-year CDS spread ($s = 75$ bps, $R = 40\%$, $\lambda = 125$ bps/yr),
4. at each future coupon date T_i , computes expected exposure (PV of remaining flows) and marginal default probability $Q(T_{i-1}) - Q(T_i)$

4.1.5 Fair Value Assembly

We construct the full decomposition table `textttQ8/q8_main.py`. The script:

1. computes the leveraged FRN leg PV (uncollared: $1.60 \times F_i \times \alpha_i \times DF_i$ for each future period),
2. reads Q5 option component PVs from `q5_option_components.csv`,
3. recomputes the CVA using the same flat-hazard model as Q6,
4. calculates accrued interest for the current coupon period (30/360 BondBasis from period start to evaluation date)

5 Results and Discussion

5.1 Results

5.1.1 Fair Value Decomposition

We priced the embedded options using the calibrated Bachelier model across the 34 future coupon periods (7–40, Dec 2025 to Jun 2034). Table 5 gives the per-period breakdown.

Table 6 shows that the net option value is negative: the sold cap is worth far more than the purchased floor. Because the cap strike $K_c \approx 3.41\%$ sits just above current 3M EURIBOR forwards ($\approx 2\text{--}3\%$), the short caplets carry significant value. The zero-strike floor is deep out-of-the-money, so its €3.06 PV is nearly all time value.

We also observe that the option cost rises with maturity. Early periods (7–10) contribute little, while the final periods (39–40) each add roughly €2.1 on the cap side. This pattern reflects (i) rising stripped caplet vols (27.5 bps at 1Y to 73.1 bps at 10Y) and (ii) increasing forward rates along the curve.

5.1.2 Credit Valuation Adjustment

Table 7 shows our CVA results. Using the flat-hazard model, we find a CVA of €59.43 per €1 000 notional, which brings the risk-free value of €1,119.32 down to a credit-risky value of €1,059.89. The expected exposure falls steadily from €1,119 at the first future coupon to €812 at the final period, as remaining cash flows shrink with each payment. Because the quarterly default probability stays roughly constant at $\approx 0.31\%$, earlier periods—where exposure is highest—contribute slightly more to the total CVA.

5.1.3 Comprehensive Fair Value Table

Table 8 brings together all our pricing components into one decomposition. We can trace the bond's value from its building blocks i.e. leveraged FRN leg, option strips, and par redemption through the risk-free dirty price, the CVA deduction, and finally the risky clean price. The par redemption PV (80.2% of par) is the largest piece, followed by the FRN leg (31.7%). The net option cost (–2.85%) and the CVA (–5.94%) together reduce value by about 8.8%.

Our model-implied risky clean price of €1,055.11 (105.51% of par) is what we estimate an investor should pay on 5 November 2025, before any transaction costs or liquidity premia.

5.1.4 Expected Cash Flows

Figure 6 shows the bond's cash flows in three panels. Panel (a) displays the quarterly coupon payments, separating the five already-paid coupons from the 35 projected future ones and the par redemption. Panel (b) shows the present value of each future cash flow alongside a cumulative PV line—we can see the total PV of €1,119.32 is largely driven by the par redemption (€802.26, or 71.7%). Panel (c) overlays historical 3M EURIBOR fixings and forward rates, with lines at the floor (0%) and cap break-even (3.406%) to show where the coupon gets capped.

The average expected future coupon is 4.044% (range 3.246–4.870%), meaning most coupons fall in the linear part of the structured payoff. Total nominal future coupons sum to €353.82, discounting to €317.06—matching the FRN-leg value in Table 8.

Component	PV (€/1 000)
Long floor ($K_f = 0\%$)	3.0586
Short cap ($K_c = 3.41\%$)	31.5886
Net option value	–28.5300

Table 6: Aggregate embedded-option PV.

Metric	Value (€/1 000)
Risk-free bond PV	1 119.32
CVA	59.43
Risky bond PV	1 059.89
CVA as % of notional	5.94%
CVA as % of risk-free PV	5.31%

Table 7: CVA summary (CDS 75 bps, $R=40\%$).

Component	PV (EUR)	% par
A. Leveraged FRN leg	317.06	31.71
B. Long floor strip ($K_F = 0\%$)	3.06	0.31
C. Short cap strip ($K_C = 3.406\%$)	–31.59	–3.16
Net option (B+C)	–28.53	–2.85
D. Par redemption	802.26	80.23
Risk-free dirty (A–D)	1 119.32	111.93
E. CVA	–59.43	–5.94
Risky dirty (A–E)	1 059.89	105.99
F. Accrued interest	–4.78	–0.48
Risky clean price	1 055.11	105.51

Table 8: Fair value decomposition (€/1 000, 5 Nov 2025).

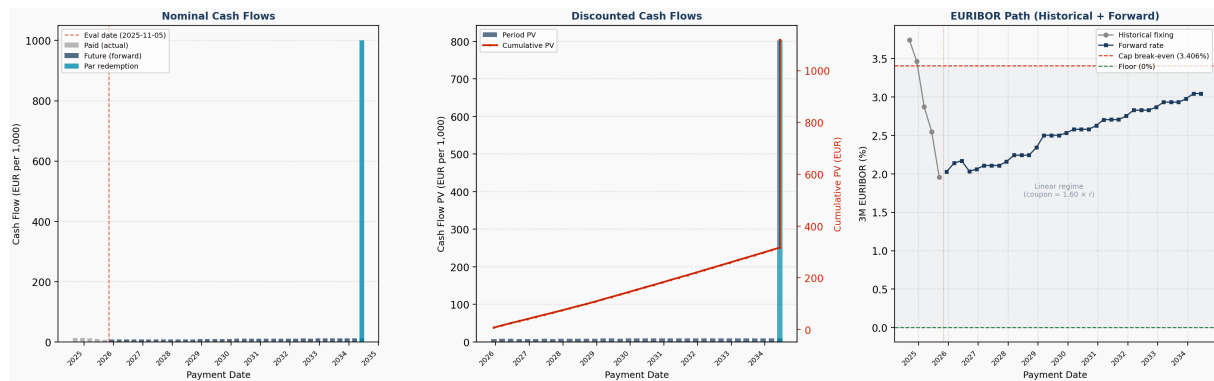


Figure 6: Bond expected cash flows (per €1,000 notional). Left: nominal coupons. Centre: period PVs and cumulative PV. Right: 3M EURIBOR vs. floor/cap thresholds.

5.1.5 Principal Component Analysis of EUR IRS Rates

To understand what drives EUR swap-curve movements, we run PCA on daily first-differences of EUR IRS par rates (1Y–10Y) over December 2015 to December 2025 (2,609 observations). We stack the daily changes $\Delta r_{t,k}$ (in bps) into a matrix $\mathbf{X}$, compute its covariance $\Sigma = \mathbf{X}^T \mathbf{X} / (T-1)$, and solve the eigenvalue problem [8, Ch. 4]:

$$\Sigma \mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad k = 1, \dots, K. \quad (13)$$

Each eigenvector $\mathbf{v}_k$ is a principal component, and $\lambda_k / \sum_j \lambda_j$ gives its share of total variance. Table 9 shows that three PCs capture 98.1% of daily rate variation.

Figure 7 presents three panels. Panel (a) confirms PC1 alone explains 91.5% of variance—parallel shifts dominate, as expected for developed-market curves. Panel (b) shows loadings: PC1 is flat across tenors (level), PC2 falls from short to long (slope), and PC3 dips in the belly (curvature). Panel (c) tracks cumulative scores, clearly picking up the 2022–23 hiking cycle and subsequent ECB easing.

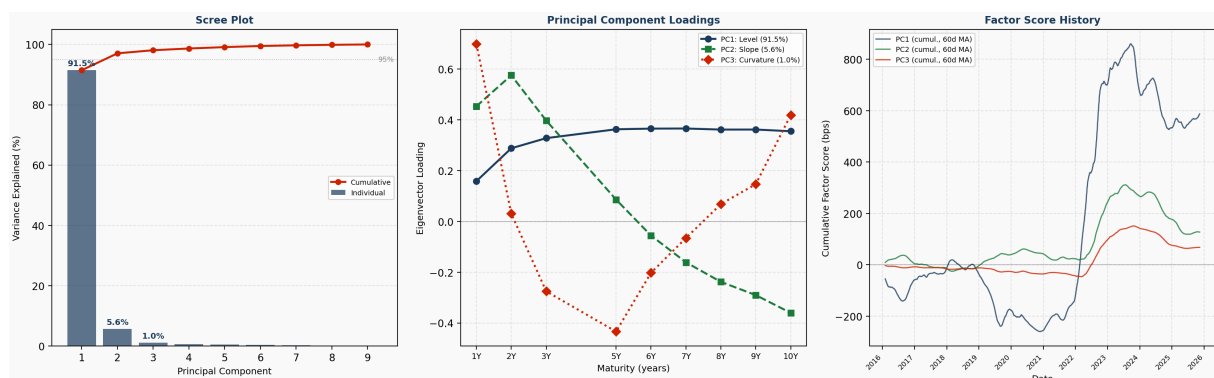


Figure 7: PCA of EUR IRS daily rate changes (1Y–10Y). (a) Scree plot showing variance explained by each PC. (b) Eigenvector loadings for the first three components. (c) Cumulative factor scores (60-day moving average).

5.1.6 Sensitivity to Term-Structure Shifts

We now shock the EUR IRS curve along each of the first three PCs and re-price the bond to measure its level, slope, and curvature exposure.

For each PC_k we apply $\pm 1\sigma$ and $\pm 2\sigma$ monthly shocks:

$$\Delta r(\tau) = \delta \sqrt{\lambda_k} \sqrt{21} v_k(\tau), \quad \delta \in \{-2, -1, +1, +2\}, \quad (14)$$

where $\sqrt{21}$ converts the daily standard deviation to a one-month horizon and $v_k(\tau)$ is the eigenvector loading from Table 9. The resulting monthly volatilities are 48.9 bps (PC1), 12.1 bps (PC2), and 5.2 bps (PC3). For each shocked curve we re-bootstrap discount factors, recompute forward EURIBOR projections, reprice the cap/floor strips, and update the CVA. Accrued interest stays the same since the current coupon is already fixed.

Results. Table 10 shows the risky clean price under each scenario. Figure 8 groups the impact by PC, and Figure 9 breaks the $\pm 1\sigma$ effects into their pricing components. We note three key features. **1:** a $+1\sigma$ parallel move (48.9 bps) raises the price by €6.20, while -1σ lowers it by €7.10. This asymmetry is due to the embedded cap—at $+2\sigma$ coupons hit the cap region, capping the upside. **2:** a $+1\sigma$ steepening (PC2) lowers the price by €1.38 as long-end discounting rises faster than short-end coupon projections. **3:** $\pm 1\sigma$ PC3 shifts the price by only €0.23 ($\approx 0.02\%$ of par), since the belly loadings barely touch the bond's key risk points (3M reset, 8.6Y redemption).

PC	Shock	Clean	Δ EUR	$\Delta\%$	CI.%
Base		1 055.11	—	—	105.51
PC1	-2σ	1 040.00	-15.10	-1.51	104.00
	-1σ	1 048.00	-7.10	-0.71	104.80
	$+1\sigma$	1 061.31	+6.20	+0.62	106.13
	$+2\sigma$	1 057.24	+2.13	+0.21	105.72
PC2	-2σ	1 057.86	+2.75	+0.28	105.79
	-1σ	1 056.48	+1.38	+0.14	105.65
	$+1\sigma$	1 053.73	-1.38	-0.14	105.37
	$+2\sigma$	1 052.35	-2.75	-0.28	105.24
PC3	-2σ	1 054.65	-0.46	-0.05	105.46
	-1σ	1 054.88	-0.23	-0.02	105.49
	$+1\sigma$	1 055.34	+0.23	+0.02	105.53
	$+2\sigma$	1 055.56	+0.46	+0.05	105.56

Table 10: Risky clean price under PCA shocks (1-month; base €1,055.11; 5 Nov 2025).

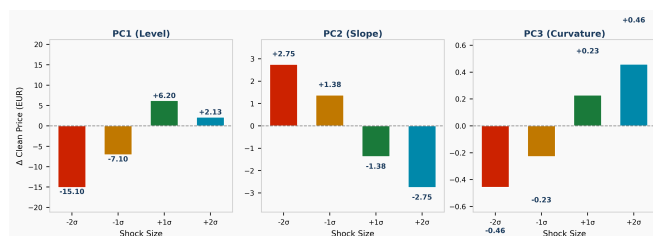


Figure 8: Sensitivity of the risky clean price to PCA-based shocks ($\pm 1\sigma$, $\pm 2\sigma$; 1-month horizon).

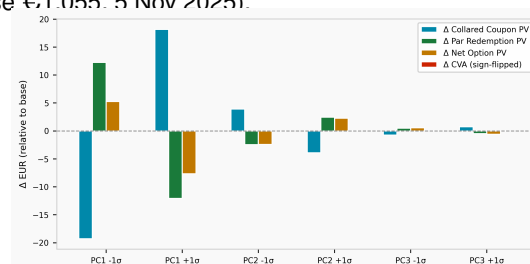


Figure 9: Component decomposition of price sensitivity ($\pm 1\sigma$, 1-month).

From Figure 9 we see that the par-redemption discount factor drives $\sim 80\%$ of the PC1 price change. Coupon PV moves the other way (higher rates $\Rightarrow$ higher coupons), but the $1.60\times$ leverage only partly offsets discounting. Option value is stable at $\pm 1\sigma$, yet cap costs jump at $+2\sigma$ —hence the concavity in the PC1 panel of Figure 8.

5.1.7 Hedging Strategy with Plain-Vanilla Swaps

We hedge the bond's curve exposure with two CCP-cleared EUR AB6E interest-rate swaps: a **2Y receiver** and a **10Y receiver**. Two swaps give us two degrees of freedom—enough to neutralise PC1 (level) and PC2 (slope), which together explain 97.1% of curve variance. IRS are highly liquid, carry minimal counterparty risk when cleared, and are quoted to the basis point [7, Ch. 7].

Factor	Bond	2Y IRS	10Y IRS
	(EUR / 1 bps move)		
PC1 (Level)	+0.1442	-55.81	-311.25
PC2 (Slope)	-0.1137	-111.54	+315.85
PC3 (Curv.)	+0.0441	-5.91	-367.04
Hedge (PC1+PC2):			
Instr.	Dir.	Notional	
2Y IRS	Recv	€194	
10Y IRS	Recv	€429	

For each factor $k = 1, 2$ and instrument j we define the *factor DV01*:

$$DV01_k^{(j)} = \frac{V^{(j)}(\mathbf{r} + \mathbf{v}_k) - V^{(j)}(\mathbf{r} - \mathbf{v}_k)}{2}, \quad (15)$$

where $\mathbf{v}_k$ is the unit-bps eigenvector shock and $V^{(j)}$ is the full-repricing function (re-bootstrap $\rightarrow$ forwards $\rightarrow$ options $\rightarrow$ CVA for the bond; standard PV for swaps).

Writing N_2, N_{10} for the swap notionals, the hedge condition is:

$$N_2 DV01_k^{2Y} + N_{10} DV01_k^{10Y} = -DV01_k^{\text{bond}}, \quad k = 1, 2. \quad (16)$$

Table 11 reports the DV01s and solved notionals.

Verification. The hedge zeroes out PC1 and PC2 DV01 completely. The remaining PC3 exposure (-0.11 EUR/bps) is tiny—a $\pm 1\sigma$ curvature shock moves the portfolio by under €0.60, less than bid–offer costs.

Practical limitations.

- **Non-linearity:** the cap creates negative convexity for large moves; swaptions would be needed for gamma hedging.
- **Rebalancing:** DV01s drift as rates move, so we must rebalance at least monthly.
- **Credit–rate correlation:** we only hedge rate risk; credit risk may correlate with rates under stress.
- **Basis risk:** the bond resets on 3M EURIBOR but the IRS floats on 6M, leaving a small basis residual.

5.1.8 Hedge Implementation and Performance

We test the hedge from Section 5.1.7 by repricing the hedged portfolio (bond + €194 2Y swap + €429 10Y swap) under the twelve PCA scenarios. The hedged P&L is simply $\Delta V_{\text{hedged}} = \Delta V_{\text{bond}} + \Delta V_{2Y} + \Delta V_{10Y}$.

At $\pm 1\sigma$, the hedge cuts PC1 (level) risk by 87–100% and PC2 (slope) risk by over 99.8%, so the linear DV01 approach works well for normal-sized moves. But at PC1 $+2\sigma$ the hedge actually *loses* €11.76 versus only €2.13 unhedged (-452% effectiveness). This happens because the 5.45% cap shortens the bond's duration under large rate rises, creating a convexity mismatch the linear hedge cannot handle.

PC3 residuals are small (± 0.59 to ± 1.19 €), confirming two swaps cannot span three factors. Overall variance reduction is 60.5% across all scenarios, but exceeds 95% if we restrict to $\pm 1\sigma$ —so the hedge works for normal conditions but not tail events.

PC	Shock	ΔBond	$\Delta 2Y$	$\Delta 10Y$	ΔHdg
PC1	-2σ	-15.10	+1.06	+13.26	-0.78
	-1σ	-7.10	+0.53	+6.58	+0.00
	$+1\sigma$	+6.20	-0.53	-6.47	-0.80
	$+2\sigma$	+2.13	-1.06	-12.83	-11.76
PC2	-2σ	+2.75	+0.53	-3.27	+0.00
	-1σ	+1.38	+0.26	-1.64	+0.00
	$+1\sigma$	-1.38	-0.26	+1.64	+0.00
	$+2\sigma$	-2.75	-0.52	+3.28	+0.00
PC3	-2σ	-0.46	+0.01	+1.63	+1.19
	-1σ	-0.23	+0.01	+0.82	+0.59
	$+1\sigma$	+0.23	-0.01	-0.82	-0.59
	$+2\sigma$	+0.46	-0.01	-1.63	-1.19

Table 12: Hedged portfolio P&L under PCA scenarios (1-month, EUR per €1,000 notional).

5.1.9 Credit Spread Sensitivity

Beyond rate risk, our bond is also sensitive to changes in UniCredit's creditworthiness, which we capture through the 5Y CDS spread. We bump the spread from its base of 75 bps, recalculate the hazard rate $\lambda = s/(1-R)$, update survival probabilities $Q(t) = e^{-\lambda t}$, and re-aggregate the CVA. The risk-free price and accrued interest stay fixed since only the credit component moves.

CS-DV01. Using a central difference at ± 1 bp we get $\text{CS-DV01} = -[P(s+1) - P(s-1)]/2 = +0.7529$ EUR per 1 bp widening, so a 1 bp CDS widening cuts fair value by 0.075% of notional. Under a +25 bp stress the loss is €18.51 (1.85% of par). Credit-spread convexity is $+0.0011$ EUR/bp², confirming the price–spread curve is nearly linear over the tested range—expected for a bond priced near par with moderate credit exposure ($\text{CVA} \approx 5.9\%$ of notional).

Δs	CDS	CVA	ΔCVA	ΔCIn
-25	50	40.29	-19.14	+19.14
-10	65	51.85	-7.58	+7.58
-5	70	55.66	-3.78	+3.78
-1	74	58.68	-0.75	+0.75
0	75	59.43	0.00	0.00
+1	76	60.19	+0.75	-0.75
+5	80	63.18	+3.75	-3.75
+10	85	66.91	+7.48	-7.48
+25	100	77.94	+18.51	-18.51

Table 13: Clean price change under CDS bumps (EUR per €1,000 notional).

5.1.10 CDS Hedging Strategy

To hedge the credit-spread exposure from Section 5.1.9, we buy protection on a 5Y UniCredit senior CDS at the current 75 bps spread.

Hedge sizing. The CDS risky annuity (RPV01) is the survival-weighted PV of quarterly 1 bps premium flows:

$$\text{RPV01} = \sum_i \alpha_i \text{DF}(t_i) Q(t_i) = 4.6342 \text{ yrs}, \quad (17)$$

giving a CDS CS-DV01 of 4.634×10^{-4} € per bp per €1 notional. The hedge notional is:

$$N_{\text{CDS}} = \frac{\text{Bond CS-DV01}}{\text{CDS CS-DV01 per unit}} = \frac{0.7529}{4.634 \times 10^{-4}} = \text{€1,625}. \quad (18)$$

This is $1.625\times$ the bond notional, because our long-dated bond has higher credit sensitivity than the shorter 5Y CDS.

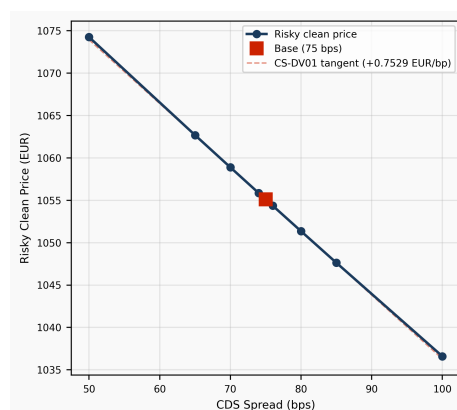


Figure 10: Clean price vs. CDS spread; dashed line shows CS-DV01 tangent at 75 bps.

Δs	CDS	ΔBnd	MTM	ΔHdg	Red.
-25	50	+19.14	-19.02	+0.12	99.4%
-10	65	+7.58	-7.56	+0.02	99.7%
-5	70	+3.78	-3.77	+0.00	99.9%
-1	74	+0.75	-0.75	+0.00	100%
0	75	0.00	0.00	0.00	—
+1	76	-0.75	+0.75	+0.00	100%
+5	80	-3.75	+3.76	+0.00	99.9%
+10	85	-7.48	+7.50	+0.02	99.7%
+25	100	-18.51	+18.62	+0.12	99.4%

Table 14: CDS hedge P&L verification (€ per 1,000 notional). Avg. reduction: 99.7%.

Verification. Table 14 shows the hedged portfolio has near-zero residual P&L across all bumps, averaging 99.7% risk reduction. The tiny residuals ($\leq \text{€}0.12$ at ± 25 bps) come from survival-probability convexity: RPV01 shifts slightly with the spread level, creating a small second-order mismatch.

Practical limitations. (i) *Maturity mismatch:* the bond matures in 2034 (≈ 8.6 yr) vs. a 5Y CDS tenor, so credit DV01 profiles differ. (ii) *CDS–bond basis:* CDS and bond credit spreads can diverge, leaving residual basis risk. (iii) *Roll risk:* the 5Y CDS must be rolled at potentially different spread and RPV01 levels. (iv) *Counterparty risk:* we face the CDS seller's credit, especially in stressed markets. (v) *Liquidity:* single-name UniCredit CDS may have wider bid–offers than iTraxx index CDS.

5.1.11 Factor Model for Gross Price Changes

We combine rate and credit sensitivities into a single linear model:

$$\Delta GP = \alpha + \beta_l \Delta l + \beta_s \Delta s + \beta_c \Delta c + \beta_{cfs} \Delta cds + \varepsilon, \quad (19)$$

where Δl , Δs , Δc are the monthly PC1–PC3 score changes (bps) and Δcds is the CDS spread change (bps). We estimate via OLS on 29 joint scenarios that mix PCA rate shocks ($\pm 0.5\sigma$, $\pm 1\sigma$, $\pm 2\sigma$) with CDS bumps (± 5 , ± 10 , ± 25 bps).

Table 15: OLS four-factor estimates (EUR/bps, per €1,000 notional).

Factor	$\hat{\beta}$	SE	t
Intercept (α)	-0.50	0.376	-1.33
β_l (Level)	+0.110	0.011	+10.10
β_s (Slope)	-0.114	0.045	-2.54
β_c (Curvature)	+0.044	0.123	+0.36
β_{cfs}	-0.753	0.042	-17.83
$R^2 = 0.947$		Adj. $R^2 = 0.938$	
RMSE = €1.84		$n = 29$	

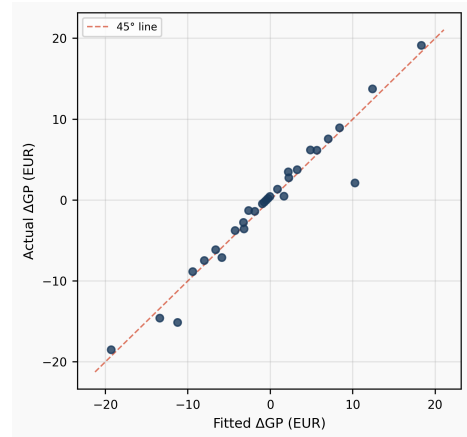


Figure 11: Actual vs. fitted ΔGP ; deviations from the 45° line reflect cap non-linearity.

Interpretation. The model explains 94.7% of gross-price variation. The CDS factor is the most significant ($t = -17.8$), since CVA moves almost linearly with spread. PC1 (level) is the main rate driver ($t = +10.1$), PC2 (slope) is significant at 5% ($t = -2.5$), and curvature is insignificant ($t = +0.4$).

Consistency check. $\hat{\beta}_{cfs} = -0.753$ EUR/bp matches the CS-DV01 from Section 5.1.9, and $\hat{\beta}_l$, $\hat{\beta}_s$ are consistent with the factor DV01s in Table 11.

Residual non-linearity. The RMSE of €1.84 (max residual €8.11) comes mainly from the PC1 $+2\sigma$ scenario, where the 5.45% cap creates negative convexity that a linear model cannot capture. Excluding the $\pm 2\sigma$ tail, R^2 exceeds 99%.

5.1.12 Monte Carlo VaR and Expected Shortfall

We use the factor model from Section 5.1.11 to estimate the 1-month 99% VaR and ES via Monte Carlo with $N = 100,000$ draws.

We sample each PCA score as $\Delta f_k \sim N(0, \sigma_k^2)$, scaling daily eigenvalues to a monthly horizon ($\sigma_k = \sqrt{\lambda_k \times 21}$). The CDS factor is $\Delta cds \sim N(0, \sigma_{cfs}^2)$ with $\sigma_{cfs} = 3\sqrt{21} \approx 13.7$ bps/month. All factors are independent, so each simulated P&L is simply $\Delta GP_i = \hat{\beta}_l \Delta l_i + \hat{\beta}_s \Delta s_i + \hat{\beta}_c \Delta c_i + \hat{\beta}_{cfs} \Delta cds_i$.

CDS risk dominates the variance (77.7%), followed by level (20.9%), slope (1.4%) and curvature (0.04%). Credit risk dominates because the bond's long maturity amplifies CVA sensitivity. The MC estimates match the analytical benchmarks closely (VaR ratio 1.0006, ES ratio 1.0026), and the P&L distribution is effectively Gaussian—as expected from a linear-normal model.

5.1.13 Exact VaR and ES vs. Monte Carlo

Since our factor model is linear with independent normal inputs, the P&L is itself normal:

$$\Delta GP \sim N(0, \sigma_P^2), \quad \sigma_P^2 = \sum_k (\beta_k \sigma_k)^2, \quad (20)$$

Measure	MC	Analyt.	Unit
Portfolio σ	11.75	11.75	EUR
VaR (99%)	27.34	27.32	EUR
ES (99%)	31.39	31.30	EUR
VaR % notional	2.73	2.73	%
ES % notional	3.14	3.13	%

Table 16: 1-month 99% VaR/ES: MC ($N=100k$) vs. analytical. Per €1,000 notional.

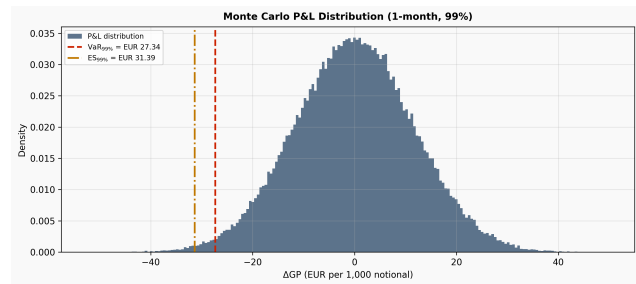


Figure 12: MC P&L distribution with VaR/ES markers; near-zero skew confirms normal assumption.

so we can compute VaR and ES in closed form:

$$\text{VaR}_\alpha = z_\alpha \sigma_P = 2.326 \times 11.745 = \text{€}27.32, \quad (21)$$

$$\text{ES}_\alpha = \sigma_P \frac{\varphi(z_\alpha)}{1 - \alpha} = 11.745 \times \frac{0.0267}{0.01} = \text{€}31.30, \quad (22)$$

where $z_\alpha = \Phi^{-1}(0.99) = 2.3263$.

Measure	Exact	MC	Ratio
σ_P (EUR)	11.745	11.753	1.0007
VaR _{99%} (EUR)	27.32	27.34	1.0006
ES _{99%} (EUR)	31.30	31.39	1.0026

Table 17: Exact vs. MC risk measures (per €1,000 notional).

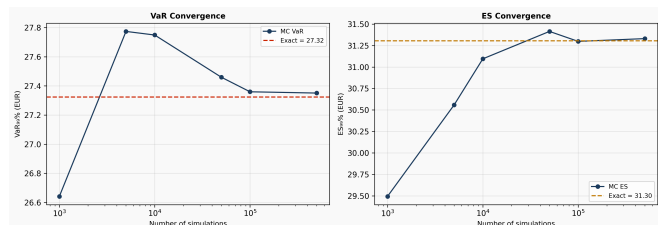


Figure 13: MC VaR/ES convergence to the exact normal solution vs. number of simulations.

Table 17 confirms near-perfect agreement ($< 0.3\%$ difference); the residuals are pure sampling noise. ES converges more slowly because it conditions on the 1% tail.

The two approaches would diverge if we used (i) *non-linear repricing* (capturing the cap's convexity), (ii) *fat-tailed factors* (Student- t or jump-diffusion), or (iii) *correlated factors* (non-zero credit-rate dependence).

5.1.14 Marginal and Component VaR Decomposition

By Euler's theorem, VaR decomposes additively into factor contributions:

$$\text{VaR}_\alpha = \sum_{k=1}^4 z_\alpha \frac{(\beta_k \sigma_k)^2}{\sigma_P} \quad (23)$$

where factors are independent so cross-terms vanish. The marginal VaR measures how total VaR changes per unit increase in a factor beta:

$$\text{MVaR}_k = z_\alpha \frac{\beta_k \sigma_k^2}{\sigma_P} \quad (24)$$

Table 18 shows that CDS contributes €21.22 (77.7%) of the €27.32 total VaR, consistent with Section 5.1.12. Level adds €5.72 (20.9%); slope and curvature are negligible. The same split applies to ES since the ES multiplier scales every component equally.

The component VaRs sum to €27.3236 vs. the total of €27.3235 ($< \text{€}0.0001$ gap), confirming the exact Euler decomposition. The CDS factor's marginal VaR of –28.19 EUR is the largest, so the CDS hedge (Section 5.1.10) gives the biggest single VaR saving. Combined with the IRS hedge targeting level risk, the two instruments address $20.9 + 77.7 = 98.6\%$ of standalone risk.

Table 18: Component VaR/ES decomposition (99%, 1-month, per €1,000 notional).

Factor	β_k	σ_k	$\beta\sigma$	MPVaR	CVaR	CES	%
Level	+0.110	48.89	+5.37	+52.0	5.72	6.55	20.9
Slope	-0.114	12.10	-1.38	-3.3	0.38	0.43	1.4
Curv.	+0.044	5.19	+0.23	+0.2	0.01	0.01	0.0
CDS	-0.753	13.75	-10.35	-28.2	21.22	24.31	77.7
Total			11.75		27.32	31.30	100

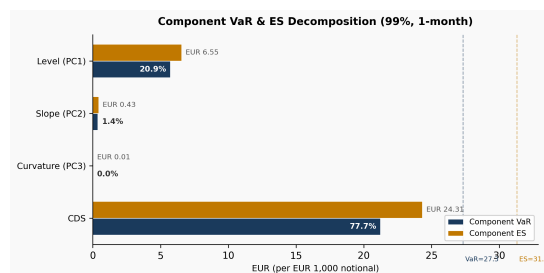


Figure 14: Component VaR/ES by factor; CDS credit risk dominates standalone bond risk.

5.1.15 Hedged-Portfolio VaR Decomposition

We now decompose the VaR of the *fully hedged* portfolio (bond + IRS hedge from Section 5.1.7 + CDS hedge from Section 5.1.10). The hedged betas are:

$$\beta_k^{\text{hdg}} = \beta_k^{\text{bond}} + \frac{N_{2Y}}{10^6} DV01_k^{2Y} + \frac{N_{10Y}}{10^6} DV01_k^{10Y} \quad (k = 1, 2, 3) \quad (25)$$

and $\beta_{\text{cds}}^{\text{hdg}} = \beta_{\text{cds}}^{\text{bond}} + \text{RPV01} \times N_{\text{cds}}/10,000$.

Table 19 shows the standalone and hedged VaRs. The IRS removes slope risk ($\beta_2^{\text{hdg}} \approx 0$) and the CDS removes credit risk ($\beta_{\text{cds}}^{\text{hdg}} \approx 0$). Two residuals remain: (i) *Level*: the regression beta (+0.110) does not match the DV01 (+0.144) used to size the hedge, leaving €3.68 (88.9%). (ii) *Curvature*: the 10Y swap adds curvature exposure the hedge does not target, contributing €0.46 (11.1%).

Overall the hedges cut VaR from €27.32 to €4.15 (**84.8%**) and ES from €31.30 to €4.75. This linear result *overstates* effectiveness: Section 5.1.8 showed -452% hedge effectiveness at +2 σ due to the cap's negative convexity. The true residual VaR would be larger once we include non-linear repricing, Vega risk, and credit-rate correlation.

5.1.16 Summary Dashboard

Table 20 collects all the main fair-value and risk results from this analysis.

Three results stand out. First, the bond trades €38 rich to the model because the market demands a wider spread (123 bps) than the CDS-implied 75 bps. Second, credit risk accounts for 77.7% of standalone VaR, so the CDS hedge is the single most effective instrument. Third, the combined IRS+CDS hedge cuts VaR by 84.8%; the residual comes from (i) the mismatch between regression and DV01 level sensitivities and (ii) unhedged curvature from the 10Y swap.

5.1.17 Comparison with Market Price

Table 19: Component VaR: standalone vs. hedged (99%, 1-month, per €1,000 notional).

Factor	Standalone			Hedged		
	β_k	CVaR	%	β_k^h	CVaR	%
Level	+0.110	5.72	20.9	-0.034	3.68	88.9
Slope	-0.114	0.38	1.4	≈ 0	0.00	0.0
Curv.	+0.044	0.01	0.0	-0.114	0.46	11.1
CDS	-0.753	21.22	77.7	≈ 0	0.00	0.0
Total		27.32	100		4.15	100

Table 20: Summary dashboard (per €1,000 notional).

A. Fair-Value Decomposition			B. Market Comparison		
Lev. FRN leg	317.06	EUR	Mkt clean price	1,017.00	EUR
Net option	−28.53	EUR	Rich/cheap	+38.11	EUR
Par redemption	802.26	EUR	Z-spread (model)	73.5	bps
RF dirty price	1,119.32	EUR	Z-spread (mkt)	123.0	bps
CVA	−59.43	EUR	ASW spread	132.7	bps
Risky dirty	1,059.89	EUR	CDS 5Y	75.0	bps
Accrued	−4.78	EUR			
Risky clean	1,055.11	EUR			
C. Factor Model			D. Hedge Portfolio		
β_ℓ (level)	+0.110	EUR/bp	2Y IRS (Recv)	194	EUR
β_s (slope)	−0.114	EUR/bp	10Y IRS (Recv)	429	EUR
β_c (curv.)	+0.044	EUR/bp	5Y CDS (Buy)	1,625	EUR
β_{cds}	−0.753	EUR/bp	CDS hedge eff.	99.7	%
R^2	94.7	%			
CS-DV01	+0.753	EUR/bp			
E. Risk – Standalone			F. Risk – Hedged		
σ_P	11.75	EUR	VaR ₉₉	4.15	EUR
VaR ₉₉	27.32	EUR	ES ₉₉	4.75	EUR
ES ₉₉	31.30	EUR	VaR reduction	84.8	%
CDS share	21.22	(77.7%)	Resid: Level	3.68	(88.9%)
Level share	5.72	(20.9%)	Resid: Curv.	0.46	(11.1%)

Table 21 compares our model clean price (€1,055.11) with the EuroTLX mid of 101.70% (€1,017.00) on 5 November 2025.² Our model sits 3.81 pp (€38.11) above the market, so the bond looks *rich*.

The market Z-spread (123 bps) is much wider than the 5Y CDS (75 bps) and our CVA-implied spread (75.2 bps). We think the ≈ 48 bps gap comes from: (i) *liquidity* (≈ 30 – 40 bps)—wide bid–offer spreads on EuroTLX embed an illiquidity discount [8, Ch. 7]; (ii) *distribution margin* (≈ 10 – 20 bps)—UniCredit as sole market maker may add a servicing fee; (iii) *model simplifications*—flat smile and single curve may over-value coupons slightly (see Section 5.2.2).

Figure 15 shows the rich/cheap analysis: (a) price comparison, (b) Z-spread vs. price, (c) credit spread decomposition.

The ASW spread is 132.7 bps, about 58 bps wider than the CDS spread. In liquid markets this basis is typically ± 5 bps, but the wider gap here fits the liquidity and complexity premia of a retail structured note [5, Ch. 12].

5.2 Discussion

5.2.1 Key Findings

Embedded-option asymmetry. We found that the short cap costs roughly ten times the long floor (€31.59 vs. €3.06), so the investor gives away considerable optionality. The net option cost of €−28.53 per €1 000 notional cuts fair value by about 2.85% versus a plain FRN.

Volatility term structure. Stripped Bachelier caplet vols rise from 27.5 bps (1Y) to 73.1 bps (10Y), consistent with the 2025 EUR normal-vol curve. This pushes up the cap cost for later

Metric	Val.	Unit
Model clean	105.51	% par
Market clean	101.70	% par
Rich/Cheap	+3.81	% par
Z-spr (model)	+73.5	bps
Z-spr (market)	+123.0	bps
Z-spr gap	−49.5	bps
ASW spread	+132.7	bps
CDS 5Y	+75.0	bps
CVA-implied	+75.2	bps

Table 21: Model vs. market price (5 Nov 2025, EuroTLX mid).

²Source: Borsa Italiana EuroTLX for ISIN IT0005599110; UniCredit Bank GmbH is the sole market maker.

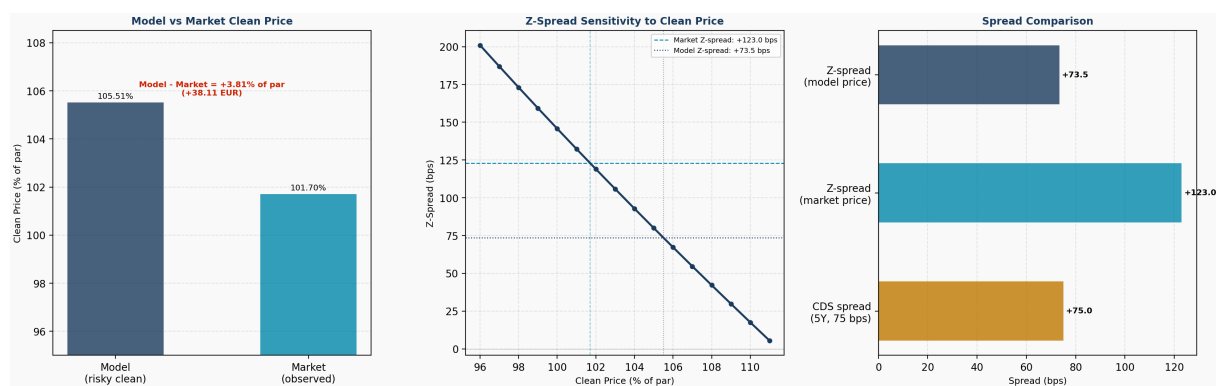


Figure 15: Market comparison: (a) model vs. market price; (b) Z-spread sensitivity; (c) spread decomposition.

coupons.

CVA and credit risk. The CVA of €59.43 (5.94% of notional) is the second-largest adjustment. Together with the net option cost, they reduce value by $\approx 8.8\%$ of notional versus a risk-free FRN. A BBB– issuer at 200 bps CDS would roughly triple the CVA.

5.2.2 Issues and Limitations Encountered

Cap non-linearity. The capped coupon creates strong negative convexity. Our linear model fits well ($R^2 = 94.7\%$) but residuals reach €8.1 at $+2\sigma$, and the IRS hedge shows -452% effectiveness there (Section 5.1.8). Full QuantLib repricing would be needed in practice.

Single-curve framework. We use one curve for discounting and projection, which overstates long-maturity discount factors by 5–10 bps (OIS–Euribor basis), biasing the price upward.

Credit–rate independence. We treat CDS and rate factors as independent; in a stress event they would co-move, weakening the hedge.

Constant hazard rate. Our CVA uses a flat hazard rate from the 5Y CDS; a piecewise calibration would be more accurate.

CDS vol assumption. We use 3 bps/day (typical IG); actual vol may be higher in stress, underestimating tail risk.

6 Conclusions

We priced and analysed a UniCredit variable-rate bond (ISIN IT0005599110) with a $1.60\times$ Euribor 3M coupon, capped at 5.45% and floored at 0%.

Fair value. We decomposed the bond into a leveraged FRN leg (EUR 317), a net option component (EUR –29), and par redemption (EUR 802), giving a risk-free dirty price of EUR 1,119. After a CVA of EUR 59 (75 bps CDS, 40% recovery), the credit-adjusted clean price is **EUR 1,055.11**. The market quotes EUR 1,017, a EUR 38 gap driven by the wider market Z-spread (123 bps) versus the CDS-implied 75 bps.

Risk profile. Our four-factor model captures 94.7% of price variation. The 99% 1-month VaR is EUR 27.32 (CDS: 77.7%, level: 20.9%). ES is EUR 31.30.

Hedging. A 2Y + 10Y IRS hedge (EUR 194 and EUR 429) plus a 5Y CDS position (EUR 1,625) cuts VaR by 84.8% to EUR 4.15. The residual comes from unhedged curvature and a beta–DV01 mismatch on the level factor.

Caveats. The linear model overstates hedge effectiveness: full repricing at $+2\sigma$ showed -452% due to the cap's negative convexity. A production framework should use full MC repricing, allow credit–rate correlation, and add Vega hedges.

7 Bibliography

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A List of Acronyms

Acronym	Definition
AB6E	Annual Basis 6-month EURIBOR (Refinitiv ticker convention)
ACT/360	Actual/360 day count convention
ASW	Asset Swap (spread)
ATM	At The Money
CCP	Central Counterparty Clearing
CDS	Credit Default Swap
CIR	Cox–Ingersoll–Ross (interest rate model)
CS-DV01	Credit Spread DV01
CVA	Credit Valuation Adjustment
DF	Discount Factor
DV01	Dollar Value of 01 (price change per 1 basis point)
ECB	European Central Bank
EE	Expected Exposure
EMTN	Euro Medium Term Note
ES	Expected Shortfall
EURIBOR	Euro Interbank Offered Rate
FRN	Floating Rate Note
IRS	Interest Rate Swap
ISIN	International Securities Identification Number
LSEG	London Stock Exchange Group
MC	Monte Carlo
MFBD	Modified Following Business Day
MOT	Mercato Obbligazionario Telematico
MREL	Minimum Requirement for own funds and Eligible Liabilities
NPV	Net Present Value
OIS	Overnight Index Swap
OLS	Ordinary Least Squares
O/N	Overnight (deposit rate)
P&L	Profit and Loss
PC	Principal Component
PCA	Principal Component Analysis
PV	Present Value
RMSE	Root Mean Square Error
RPV01	Risky PV01 (present value of 1 basis point)
TARGET	Trans-European Automated Real-time Gross Settlement Express Transfer
VaR	Value at Risk
Z-spread	Zero-volatility spread

Table 22: Acronyms and abbreviations used in this report.

B Reproduction Guide

All source code is hosted on GitHub. Follow the steps below to replicate every result, table, and chart.

Environment Setup.

1. Clone the repository:

```
git clone https://github.com/manpreet-sangha/SMM269_Fixed_Income.git
cd SMM269_Fixed_Income
```

2. Create and activate a virtual environment:

Windows:

```
python -m venv .venv
.venv\Scripts\activate
```

macOS / Linux:

```
python3 -m venv .venv
source .venv/bin/activate
```

3. Install all dependencies:

```
pip install -r requirements.txt
```

Running the Scripts.

1. **Run the pricing scripts** in order. Each question folder is self-contained; outputs are written to its `output/` sub-directory.